

B+ Tree Range Scan

B+ Trees, Leaf-Linked Index, Range Query · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★★★★ Advanced

Problem

Pattern: leaf-linked multiway index with a range scan. Bulk-load sorted keys into the leaves of a B+ tree (each leaf holds up to L keys), derive the separator keys, and answer a range query by walking the leaf chain.

Return the number of leaves, the separator (index) keys, the leaf chain, and the keys within the query range.

What to Compute

- Leaves: the number of leaf nodes.
- Index Keys: the first key of every leaf after the first (the separators a parent would store).
- Leaf Chain: each leaf's keys; and the keys k with $lo \leq k \leq hi$ for the range query.

Input / Output Format

Input: Line 1: L (leaf capacity). Line 2: N. Line 3: N integer keys. Line 4: lo hi.

Output: Four lines: Leaves, Index Keys, Leaf Chain, Range [lo,hi].

Examples

Example 1

Input: 3 9 1 2 3 5 8 13 21 34 55 4 22

Output:

Leaves: 3
Index Keys: 5 21
Leaf Chain: 1,2,3 5,8,13 21,34,55
Range [4,22]: 5 8 13 21

Explanation: Nine sorted keys pack into three leaves of three; the separators 5 and 21 route searches, and scanning the chain yields the keys between 4 and 22.

Example 2

Input: 2 6 1 2 3 4 5 6 2 5

Output:

Leaves: 3
Index Keys: 3 5
Leaf Chain: 1,2 3,4 5,6
Range [2,5]: 2 3 4 5

Explanation: Six keys pack into three leaves of two; separators 3 and 5 route searches, and the range [2,5] collects 2,3,4,5 from the chain.

Constraints

- De-duplicate and sort the keys, then pack them left to right into leaves of up to L keys.
- A separator is the first key of each leaf except the first.
- The range scan returns in-range keys in sorted order.

Sample Run

Sample Input

3
9
1 2 3 5 8 13 21 34 55
4 22

Sample Output

Leaves: 3
Index Keys: 5 21
Leaf Chain: 1,2,3 5,8,13 21,34,55
Range [4,22]: 5 8 13 21